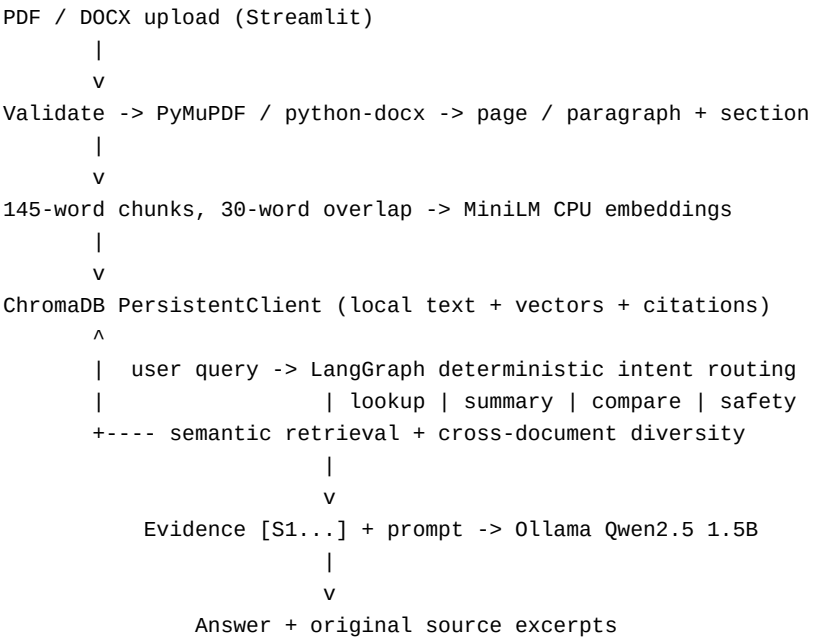
Clinical Research AI Copilot

Goal. A free, local-first, source-grounded document copilot for the interview assignment; it supports multiple PDF and DOCX files, semantic retrieval, cross-document analysis and traceable references.

1. Component architecture



2. Retrieval and generation flow

Ingestion. Verify file header/type and 15 MB limit, extract text, retain genuine PDF page or DOCX paragraph, group by section, chunk to fit MiniLM, embed locally and persist to Chroma. SHA-256 deduplicates uploads; reupload by filename replaces older chunks.

Question. The LangGraph router chooses lookup, summary, compare or safety. Chroma cosine similarity retrieves chunks; compare/safety add query expansions and document diversity. The generator receives only retrieved excerpts, each tagged with a deterministic source ID. UI displays the excerpt and metadata.

3. Technology choices

Python 3.11 + Streamlit avoids a separate JS frontend. **Ollama qwen2.5:1.5b** is a small local model better suited to 8 GB RAM than 7B defaults. **all-MiniLM-L6-v2** creates local vectors; **ChromaDB** persists them without a hosted service. **LangGraph** makes routing visible and testable.

Design Decisions & Guardrails

4. Grounding and citations

Every retrieved chunk carries filename, PDF page number (where applicable), DOCX paragraph location and heuristic section. The model is told to cite [S#] evidence IDs; unknown IDs are stripped in code. Source excerpts always come from Chroma metadata, not generated page guesses. Similarity is a retrieval heuristic, not a correctness probability.

5. Multi-document reasoning

For safety/compare queries, retrieve several semantic variants, then reserve relevant evidence across distinct documents. Example: compare protocol section 6.2 with monitoring finding MV-01 and deviation DEV-001. Separate observed noncompliance from potential clinical implications; do not assert actual harm. Protocol v1/v2 comparison highlights the v2 Week 8 monitoring addition.

6. Security and operational stance

Local-only model inference avoids paid API dependencies, but Chroma persists plaintext snippets on disk. No authentication, encryption-at-rest, user isolation, PHI support, or validated regulated audit trail is provided. Do not host this public or upload patient-identifiable records. Demo files are entirely synthetic. PDF OCR and complex table fidelity are not implemented.

7. Testing and evaluation

Pytest checks chunk boundaries, routing, uploads, citations and parsing. Retrieval tests check expected source documents on synthetic cases. During the demo inspect both answer and cited excerpts manually, including an unsupported question. Evaluate retrieval relevance, citation correctness and refusal behavior separately; the small model can still hallucinate.

8. Tradeoffs / planned enhancements

The rule-based intent router is transparent and dependable but is not a fully autonomous LLM agent. Additional work: study-level filtering, richer PDF/Word table extraction, OCR, lexical+vector hybrid search, measured retrieval thresholds, stronger LLM if hardware permits, human-validated answer grading, access control, audit logging and versioned prompt/model governance.

9. Free reproducibility

Installation: Python 3.11, local Ollama with `qwen2.5:1.5b`, `pip install -r requirements.txt`, and Streamlit. One-time internet is needed for package and model downloads. No paid service or API key is required.